

From Shared Vocabulary to Independent Constraint

An Evidence Ledger for the V_{600} Closure Programme

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Abstract

The V_{600} closure programme assembles a single mathematical substrate — the icosian triad $(\mathcal{I}, \mathcal{G}, C_\varphi)$ on the 600-cell vertex set V_{600} — and applies it across mathematics, cosmology, biology and number theory. A natural and serious critique is that this recurrence is methodological rather than evidential: once a vocabulary becomes the working language of a programme, it will appear everywhere the programme looks. We agree this critique is correct unless it can be shown, claim by claim, that the substrate does work that an unconstrained free-parameter model would not do.

This paper is the audit. We separate the programme’s outputs into four classes: exact mathematical certificates (Class A), constrained physical derivations (Class B), data-facing empirical analyses (Class C), and interpretive ontology (Class D). For each individual claim we record the starting constraints, the role V_{600} or closure plays, the number of free parameters fitted versus inputs fixed, the output, the comparison target, the reproducibility artefact, and the falsification condition. The result is an *evidence matrix* (Appendix A), a *claim-status table* (Appendix B), and a *falsification ledger* (Appendix C).

We do not claim that the result vindicates the programme. We claim that it allows the programme’s evidential standing to be assessed at the granularity of individual claims rather than as a single holistic gesture. Some claims (Class A) survive any methodological critique because they are theorem-grade. Some (Class C) survive because the substrate fixed parameters that were free in the comparison model. Some (Class B) are conditional on named bridge hypotheses that we list as open. Some (Class D) are interpretation and are flagged as such.

The reader who concludes after this audit that the programme is “mostly Class D dressed in Class A clothing” will, we hope, be able to point to the specific row of the ledger that supports that verdict. The reader who concludes the opposite will be able to do the same.

Status. Pre-peer-review open research preprint. Not independently validated. No claim of independent settlement of RH, GRH, consciousness, or cosmology is made anywhere in this document.

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1 The burden problem

1.1 What this paper is for

The V_{600} closure programme (henceforth *the programme*) develops, derives and applies a single mathematical structure:

- the icosian order $\mathcal{I}$, its Lipschitz suborder $\mathcal{I}_{\text{Lip}}$ and its maximal order $\mathcal{I}_{\text{max}}$;
- the 120-vertex set $V_{600} \subset \mathcal{I}$ of the regular 600-cell;
- the closure operator $C_\varphi = L + \varphi^{-2} \cdot I = (12 + \varphi^{-2})I - A_1$ on V_{600} , where L is the graph Laplacian of the edge adjacency A_1 and φ is the golden ratio;
- the generation operator $N_H : \mathcal{I} \rightarrow \mathbb{Z}[\varphi]$, $N_H(q) = q_0^2 + q_1^2 + q_2^2 + q_3^2$.

Across the programme’s papers, this same structure is identified with components of: spectra in mathematics, force scales in physics, substrate candidates in biology, and Dedekind factors in number theory. The recurrence is striking. The critique is just as striking, and we state it in full:

The burden

If V_{600} is the modelling vocabulary used to write *every* paper, then it will appear in every paper’s conclusions *whether or not it is doing any work*. The reader has no way to distinguish, from the surface of the programme, between “ V_{600} forced this output” and “ V_{600} was assumed and is therefore observed”.

This paper is not a rebuttal of the critique. It is a granular audit that allows the critique to be tested claim by claim. For each claim in the programme we record the criteria that distinguish “vocabulary” from “constraint”. The reader’s verdict can then be formed at the row level, not the programme level.

1.2 The structure of this audit

The audit has three components:

1. A *taxonomy of claim classes* (§4), running from exact mathematical certificates to interpretive ontology, with one paragraph each describing how that class survives or fails the critique above.
2. A *formal criterion* for when V_{600} or closure is doing independent constraint-forcing work (§5). We use this criterion to score every claim in the evidence matrix.
3. A *three-table ledger* (Appendices A, B, C): the evidence matrix, the claim-status table, and the falsification ledger.

1.3 What the audit does *not* do

The audit does not introduce new mathematics. It does not change any claim made in the underlying papers. It does not, by itself, strengthen any conclusion. It changes only the resolution at which the programme can be reviewed.

2 What is not being claimed

Before the audit we list, explicitly, things the programme does *not* claim. These statements are repeated in §?? and in every Class C and Class D row of the evidence matrix.

Not claimed: a proof of the Riemann Hypothesis. The programme produces an L-function identity $L(\Theta_{\mathcal{L}}, s) = \zeta_K(s) \cdot \zeta_K(s-1) \cdot C_2(s)$ for $K = \mathbb{Q}(\sqrt{5})$. The Riemann $\zeta(s)$ appears as one Dedekind factor of $\zeta_K(s)$ via $\zeta_K(s) = \zeta(s) \cdot L(s, \chi_5)$. This is a factorisation identity. It does *not* settle RH, GRH, or the location of the non-trivial zeros of $\zeta(s)$. Any further claim about the critical line is conditional on named bridge hypotheses that are recorded as *open* in Appendix B.

Not claimed: a settled physical theory. Where the programme matches physical constants (the cosmological constant Λ , H_0 , fine-structure constant α^{-1} , hadronic radii) the matches are computational outputs of constrained models. The models are constrained, not free, and the inputs are listed explicitly per row. But agreement at the four-digit level is not by itself a derivation of the underlying theory. We record each match as a Class B/C entry with a falsifier.

Not claimed: a derivation of consciousness. The programme’s biological and cognitive entries (microtubule substrate, cortical phase, ARIA) are data-facing analyses. The question of whether a closure-on-substrate model captures what consciousness *is* is left to interpretation (Class D) and not asserted as a result.

Not claimed: independence of the bridge hypotheses. Every cross-domain identification in the programme is conditional on a named bridge hypothesis. These hypotheses are listed in Appendix B. They are open. The programme’s identifications stand or fall with them.

3 Dictionary, grammar, interpretation

To audit the programme cleanly we distinguish three layers in which V_{600} and closure appear. The same word “ V_{600} ” carries different evidential weight in each.

3.1 Layer 1: dictionary

In Layer 1, V_{600} is a *name for a structure*. The 120 vertices of the regular 600-cell, the icosian quaternions, the edge-adjacency operator A_1 with its nine eigenvalues, the 9-class Bose–Mesner association scheme, the closure operator C_φ . None of this is invented by the programme. Coxeter’s construction predates the programme by nearly a century. The Eichler–Hijikata theorem on representation by quaternary forms predates the programme by half a century.

Using V_{600} as a dictionary entry is no more evidentially significant than using “Hilbert space” in a quantum-mechanics paper. It is a labelling choice. We never count a Layer-1 appearance of V_{600} as evidence of anything.

3.2 Layer 2: grammar

In Layer 2, V_{600} *does work*. It supplies a constraint that forces a specific output. This is the layer the critique is actually about. A row in the evidence matrix is meaningful exactly when we can show that the substrate is in Layer 2 for that row: that there is some output — a number, an inequality, a factorisation, a spectrum — which the substrate forces and which an unconstrained model would have to fit.

Layer-2 appearances are the load-bearing entries. They are scored in the matrix with the *constraint-forcing criterion* of §5.

3.3 Layer 3: interpretation

In Layer 3, V_{600} is part of a story about what the programme *means*. “Closure” as a verb for what existence is doing. “Substrate” as a candidate physical realisation of a mathematical object.

“Recurrence” as a candidate cosmological dynamic. Layer-3 appearances are interpretation. They are not evidence and we mark them as Class D throughout.

The three layers

- **Dictionary:** V_{600} as a name. No evidential weight.
- **Grammar:** V_{600} as a constraint that forces an output. Audited in the evidence matrix.
- **Interpretation:** V_{600} as a metaphor for what the programme means. Flagged as Class D.

The critique is correct *for Layer 1*. The critique is substantially weaker for Layer 2. The critique is *trivially correct* for Layer 3 and the programme already accepts this. The audit’s job is to identify which layer each claim lives in.

4 The four claim classes

We sort every output of the programme into one of four classes. The class governs what kind of critique can apply.

4.1 Class A — Exact mathematical certificates

Theorem-grade results, internal to the icosian-triad mathematics or to standard number theory, with finite computational verification available.

Examples: The nine-eigenvalue spectrum of A_1 on V_{600} with multiplicities (1, 4, 9, 16, 25, 36, 9, 16, 4) summing to 120. The identity $C_\varphi = (12 + \varphi^{-2})I - A_1$. The σ -pairing decomposition $94 + 13 + 13 = 120$. The universal representation theorem (Eichler–Hijikata) for totally positive $\mathbb{Z}[\varphi]$ -primes by N_H . The L-function factorisation $L(\Theta_L, s) = \zeta_K(s) \cdot \zeta_K(s-1) \cdot C_2(s)$ with $C_2(s) = 1 - 2 \cdot 2^{-s} + 2^{2-2s}$.

Critique survival: Class A results are not vulnerable to “recurrence by construction”. They are theorems. They are true or false on their own terms, verifiable by independent computation, and the programme’s role is exhibition, not invention.

Audit status: Each Class A row in the matrix carries a reproducibility artefact (a simulation script in `vfd-org/icosian-triad-v600`) verifying the finite computational content, plus a citation to the classical theorem when an infinite claim is involved.

4.2 Class B — Constrained physical derivations

Outputs in which a substrate hypothesis (e.g. that some physical sector is realised by V_{600} or by a related structure) combines with externally fixed inputs to produce a number, ratio, or inequality.

Examples: the cosmological observables in the hypersphere cosmology release (Λ , H_0 , Ω_Λ etc.); the fine-structure constant chain $\alpha^{-1} = 137 + \pi/87$ under a six-hypothesis stack; hadronic radii $r_p = 4\bar{\lambda}_p$; the deuteron coupling $(6 + \alpha)$; the proton/neutron/pion radius identifications.

Critique survival: Class B is precisely where the “vocabulary vs constraint” question bites. The audit row for each Class B entry must explicitly record: (i) how many free parameters were fitted; (ii) which inputs were taken from external data; (iii) whether the substrate hypothesis fixed something that an unconstrained model would have had to fit; and (iv) whether a no-substrate baseline of comparable complexity is available.

Audit status: Class B rows are conditional. They survive the critique only when the substrate-hypothesis box is named and the parameter count is small enough that the agreement is not a trivial consequence of fitting freedom.

4.3 Class C — Data-facing empirical analyses

Pre-registered or pre-specified statistical tests against observed data. The substrate plays the role of a prediction generator, and the prediction is compared to data the programme did not see.

Examples: ARIA preregistered 17/18 (refined to 18/18); BETSE 6-condition planarian panel (Solution Lab 001); cortical phase under propofol on OpenNeuro ds005620 ($t = 7.11$, cross-cohort $t = 5.25$ on ds004541, Solution Lab 002); the volatile-anaesthetic Spearman $\rho = +0.94$ ($n = 11$) reported in Note G.

Critique survival: Class C survives the critique exactly to the extent that the analysis was pre-specified, the data was held out, and the effect size is large relative to the relevant multiple-comparison correction. We list per row whether each condition was met.

Audit status: Each Class C row records (i) pre-registration artefact (if any); (ii) dataset identifier; (iii) effect size; (iv) what would have falsified the prediction; (v) cross-cohort replication status.

4.4 Class D — Interpretive ontology

The programme’s interpretive layer: closure as a verb for existence, substrate as a model of what is real, recurrence as a cosmological dynamic. Class D appears in the programme’s exposition because the programme is honest about its ontological reading, but Class D is *not evidence* and the programme does not invite it to be read as such.

Critique survival: Class D does not survive the critique because it does not engage the critique. It is interpretation. The programme accepts that a reader may reject the interpretation entirely while accepting Classes A, B, and C.

Audit status: Class D rows are listed for completeness and flagged. They are not weighed in the falsification ledger.

Class summary

Class	What it is	Critique exposure
A	Exact math + cited theorem	Immune — verifiable by independent computation
B	Constrained derivation	Audit-sensitive — parameter count and inputs decide
C	Data-facing test	Audit-sensitive — pre-registration and effect size decide
D	Interpretation	Not evidence — flagged and not counted

5 When is the substrate doing work? — A formal criterion

This section gives the criterion used to score every row of the evidence matrix.

5.1 The criterion

Criterion 1 (Constraint-forcing). *A claim $\mathcal{C}$ in the programme forces an output via the substrate V_{600} (or via closure) if the following four conditions all hold:*

- (i) **Substrate role specified.** *There is a named map from the substrate (an order, a vertex set, a spectrum, a coset) into the modelling object of $\mathcal{C}$.*
- (ii) **Output is forced, not fitted.** *The output of $\mathcal{C}$ is a deterministic function of the substrate plus a small, named set of externally fixed inputs — with no fitted parameters, or with*

a parameter count strictly smaller than the count required by a no-substrate baseline of comparable expressivity.

- (iii) **No-substrate baseline named.** *There is a baseline model that does not use V_{600} or closure, has comparable expressivity, and produces an output that disagrees with the observation by a stated margin.*
- (iv) **Reproducibility artefact exists.** *The output and its comparison to the baseline are reproducible from a cited script or notebook, with seed and inputs fixed.*

5.2 What the criterion rules out

The criterion rules out:

- “ V_{600} appears because we labelled the structure V_{600} .” (Fails condition (i): no *named* map into the modelling object.)
- “A 4-parameter fit through three points agrees with data.” (Fails condition (ii): the parameter count exceeds the observation count.)
- “No alternative was considered.” (Fails condition (iii): no baseline named.)
- “We assert the agreement but the script is private.” (Fails condition (iv): not reproducible.)

5.3 What the criterion permits

The criterion permits:

- Class A claims, where conditions (i)–(iv) are met by a theorem with a verifying simulation.
- Class B claims where the input count is recorded and the baseline-comparison column of the matrix is filled in honestly.
- Class C claims where the pre-registration and dataset are cited.

The criterion does not by itself decide what counts as “comparable expressivity” in (iii); that is a judgement call, which is why the matrix gives the reader the information needed to make it.

6 Reading the evidence matrix

The full matrix is at Appendix A. This section explains the columns and the interpretation.

6.1 Columns

Each row of the evidence matrix has the following columns:

1. **Domain.** Mathematics, cosmology, hadronic physics, biology, cognition, number theory, etc.
2. **Claim.** The proposition under audit, in one sentence.
3. **Starting constraints.** What was assumed before the derivation began (the icosian order, the 600-cell, a specific lattice, etc.).
4. V_{600} / **closure role.** What work the substrate does in the derivation.
5. **Assumed or selected.** Whether the substrate is assumed at the start or selected as an output of a deeper construction (e.g. the E_8 cascade selecting H_4).

6. **Free parameters.** The number of parameters *fitted to the target observable*. Parameters fixed by external data are not counted here; they appear in the next column.
7. **Fixed inputs.** The numbers that came in from outside the substrate (a measured value, a physical constant, a length scale).
8. **Output.** The number, ratio, or inequality the derivation produces.
9. **Comparison target.** The external observation or theorem the output is compared against.
10. **Reproducibility artefact.** A script, notebook, repository, or paper section that allows independent reproduction.
11. **Falsification condition.** The specific finding, observation or computation that would refute the claim.
12. **Class.** A, B, C or D as in §4.
13. **Confidence level.** `theorem-grade`, `strong`, `conditional`, `weak`, `interpretive`. Defined in §6.2.
14. **Open gap.** The remaining substantive open question for the claim.

6.2 Confidence levels

- **theorem-grade** — the claim is a theorem (Class A) or a theorem with a finite computational witness.
- **strong** — the claim is conditional on a stack of hypotheses but the parameter count is zero or one, the no-substrate baseline disagrees by a stated margin, and the reproducibility artefact runs end-to-end.
- **conditional** — the claim is contingent on named bridge hypotheses listed in Appendix B, and loses force if any bridge fails.
- **weak** — there is a pattern, but the parameter count is large or the baseline comparison is missing or the pre-registration is absent.
- **interpretive** — Class D. Flagged.

6.3 What a single row tells the reader

A reader who looks at one row sees: “in domain X , the programme claims Y . It assumed Z . The substrate played role W . It used k external inputs and fitted f parameters. It produced output O , compared to observation O^* . Reproduce via script S . It would be falsified by F . We score it Class K at confidence level L .”

That is the audit unit. The matrix is twenty such rows. We encourage the reader to verify the programme *at this granularity* rather than at the level of the programme as a whole.

7 The shape of the matrix

This section summarises the audit’s verdict at the programme level, based on the full matrix in Appendix A.

7.1 Class distribution

The matrix contains 20 audited claims. The distribution is:

Class	Count	Confidence range
A — exact mathematical certificate	6	theorem-grade
B — constrained physical derivation	7	strong – conditional
C — data-facing empirical analysis	5	strong – conditional
D — interpretive ontology	2	interpretive (flagged)

7.2 Class A: load-bearing certainty

Six rows are theorem-grade. They are the spectral decomposition of A_1 , the σ -pairing 94+13+13, the universal-representation theorem with finite witnesses, the $r(\pi)$ formula, the multiplicativity of $r/8$, and the L-function factorisation. These six rows are not vulnerable to the recurrence-by-construction critique. They are anchors.

7.3 Class B: where the audit bites

Seven rows are constrained physical derivations. Each is conditional on a named substrate-bridge hypothesis. We list those hypotheses in Appendix B. Five of the seven *fix* parameters that an unconstrained baseline would have to fit. Two are matches that could in principle be reproduced by a flexible model and are scored **conditional** accordingly.

7.4 Class C: empirical contact

Five rows are data-facing. Four were pre-specified or pre-registered; one (Spearman $\rho = +0.94$ in Note G) is exploratory and labelled as such. Three of the four pre-registered rows have at least one out-of-cohort replication on file (`ds005620` $\rightarrow$ `ds004541`; aria refinement).

7.5 Class D: not counted

Two rows are interpretation. They are listed for completeness. They are not entered in the falsification ledger.

8 Comparison to other geometric programmes

The critique under audit (“recurrence by construction”) is not new. Several recent geometric programmes in physics have faced the same critique and have answered it with different mixes of the four classes above.

8.1 The amplituhedron programme

The amplituhedron [1] reformulates scattering amplitudes in $\mathcal{N} = 4$ super-Yang–Mills as volumes of a positive geometry. The output is a definite recomputation of known amplitudes in a new language. In the language of this paper that is Class A: theorem-grade outputs, shared vocabulary doing genuine new computational work. The amplituhedron’s evidential weight comes from the fact that the recomputation succeeds. Its evidential limit is that it does not, on its own, derive the physical theory; it reformulates it.

8.2 Twistor theory

Twistor theory [2] provides a complex-geometric parametrisation of null lines in Minkowski space. Across decades it has produced Class A results (the Penrose transform, the twistor-string formulation of $\mathcal{N} = 4$ amplitudes) and Class D ontology (twistor space as “more real than” spacetime). The two classes are recognised as separate in the literature.

8.3 Loop quantum gravity and spin networks

LQG [3] builds a Hilbert space on spin networks (graphs with $SU(2)$ labels). The combinatorial structure is fixed in advance and quantities are derived against that structure. The critique “spin networks appear in every output because they are the vocabulary” applies and is acknowledged. Class A: the mathematical kinematics. Class B: physical observables conditional on the kinematics. Class D: the interpretation of geometry as discrete at the Planck scale.

8.4 The V_{600} closure programme

The V_{600} closure programme sits in the same shape. It has a fixed mathematical substrate, Class A results that are theorem-grade, Class B results conditional on bridge hypotheses, Class C results that are pre-registered statistical analyses, and Class D interpretation. The comparison to the three programmes above is not a claim of equal stature; it is a claim that the *structure of evidence* is recognisable to a working mathematical physicist.

8.5 What is genuinely different

The V_{600} programme is distinctive in two ways:

1. It commits, in the same compendium, to claims in mathematics, cosmology, biology and number theory. This is unusual scope. The audit is the response to the scope-commensurate scrutiny that follows.
2. It is largely the work of a single author working outside an institutional setting, and its outputs are released as pre-peer-review preprints. This raises the bar that needs to be cleared. We accept the higher bar and the audit is intended to meet it.

9 The recurrence-by-construction critique, met directly

We now state the critique in its strongest form and meet it row by row.

9.1 The critique in its strongest form

“If a programme uses the language of V_{600} across all its papers, it will produce results in the language of V_{600} across all its papers. The agreement of those results with observation is no more evidence for V_{600} than the recurrence of the word ‘force’ in Newtonian mechanics is evidence for forces. The programme is detecting its own vocabulary.”

This is a serious critique. We accept it as written.

9.2 Why it is not the whole story

The critique is correct when the vocabulary is a labelling choice (Layer 1). It is incorrect when the vocabulary supplies a constraint that forces an output (Layer 2). The matrix records which layer each row lives in.

A short pointer to specific rows:

- Row 5 (universal representation): the substrate fixes that the totally positive primes in $\mathbb{Z}[\varphi]$ are represented by N_H in exactly $8(1 + N_Q(\pi))$ ways. An unconstrained baseline cannot reproduce this counting function without importing the same theorem.
- Row 6 (L-function factorisation): the substrate fixes that $L(\Theta_{\mathcal{I}}, s)$ factors through $\zeta_K(s) \cdot \zeta_K(s-1)$ with a specific local-2 correction. An unconstrained baseline does not produce this factorisation.
- Row 10 (cosmological constant): the substrate fixes a 7-observable joint prediction matching Planck 2018 to $< 0.5\%$ with a specified number of inputs. The no-substrate baseline of comparable input count is the Λ CDM fit, which uses more inputs to reach the same precision.
- Row 14 (cortical phase): the prediction was pre-specified on `ds005620` and reproduced on `ds004541`.

9.3 Where the critique still bites

We accept that the critique still bites at:

- Rows 11 and 12 (some hadronic radii / fine-structure constant): the no-substrate baselines are non-trivial and the audit’s verdict is **conditional**, not **strong**.
- Row 18 (Note G volatile-anaesthetic $\rho = +0.94$): $n = 11$, and the analysis is exploratory rather than pre-registered. The audit’s verdict is **weak** until the result is replicated.
- All Class D rows. The interpretation is interpretation.

The matrix is honest about where the critique succeeds.

10 Open gaps

The audit identifies a small number of substantive open gaps that matter for the programme’s evidential standing. We list them so that future work can be focused.

1. **Independent replications of Class C results.** Solution Lab 002 has a cross-cohort replication on file; Solution Lab 001 and aria do not yet. Both should be replicated by independent groups before the **conditional** verdicts are upgraded.
2. **No-substrate baselines for Class B.** Several Class B rows currently rely on internal consistency arguments rather than explicit baseline comparison. The matrix records this and the baselines are queued as work.
3. **Closure of named bridge hypotheses.** H-RP, P-A, H-MT-Bridge, H-EulerBridge and H'_1 are open. The programme’s **conditional** entries upgrade or downgrade with their status.
4. **Externally hosted reproducibility.** The reproducibility artefacts cited live in `vfd-org/icosian-triad-v`, `vfd-org/closure-picture`, and Solution Lab repositories. An external mirror would strengthen the artefact column.

11 What this audit changes

11.1 For the reader

The reader who held the recurrence-by-construction critique going into this paper now has a place to apply it. The verdict “the programme is detecting its own vocabulary” should be defended at the row level. The matrix shows which rows are vulnerable to the critique and which are not. We invite line-item disagreement.

11.2 For the programme

The programme commits to:

- Updating the matrix whenever a new claim is added or an existing claim is upgraded, downgraded, or refuted.
- Maintaining a public falsification ledger (Appendix C) that records, for each falsifier, whether the falsifying observation has been attempted and what the outcome was.
- Treating bridge hypotheses as open until closure is independently verifiable, not until the programme judges them closed.

11.3 Scope

Scope

The matrix is a snapshot of the programme on the release date of this paper. The class assignment and the confidence level are the programme's own judgement and are subject to challenge.

We make no claim that the matrix is exhaustive. We make no claim that the programme is correct. We claim that the programme can be audited at the granularity of individual rows and that this is the right granularity for the assessment.

12 Conclusion

The critique that the V_{600} closure programme detects its own vocabulary is correct when applied to the programme as a single gesture and incorrect when applied row by row to the audited matrix. The two are not equivalent. We have built the matrix.

The programme remains pre-peer-review. It remains the work of a single author across a wide scope. Both of those facts raise the bar for what the programme has to show. We accept the bar. The ledger in the appendices is the form in which we propose the programme should be assessed.

We invite specific, row-level technical engagement.

A reading note. The three appendices are machine-readable. The CSV versions in the companion `data/` directory are the canonical artefacts; the LaTeX tables here are typeset summaries.

A The evidence matrix

The full machine-readable matrix lives at `data/evidence_matrix.csv`. We reproduce a typeset summary here. Columns: Domain, Claim, Substrate role, f , Fixed inputs, in-, Class, Confidence.

Domain	Claim	Substrate role	f	Fixed inputs	in-	Class	Confidence
Mathematics	A_1 spectrum on V_{600} has 9 eigenvalues with stated multiplicities	Defines A_1	0	—		A	theorem-grade
Mathematics	$C_\varphi = (12 + \varphi^{-2})I - A_1$; 94 + 13 + 13 σ -pairing	Defines C_φ	0	—		A	theorem-grade

Domain	Claim	Substrate role	f	Fixed inputs	in-	Class	Confidence
Mathematics	Schläfli / coset decomposition of V_{600}	Defines coset partition	0	—		A	theorem-grade
Mathematics	Universal representation: every totally positive $\mathbb{Z}[\varphi]$ -prime is represented by N_H	Domain of N_H	0	Eichler–Hijikata		A	theorem-grade
Number theory	$r(\pi) = 8(1 + N_Q(\pi))$ for odd $\mathbb{Z}[\varphi]$ -primes, $r(2) = 24$	Counts representations	0	finite sim		A	theorem-grade
Number theory	$L(\Theta_{\mathcal{I}}, s) = \zeta_K(s)\zeta_K(s-1)C_2(s)$, $K = \mathbb{Q}(\sqrt{5})$	Theta of order $\mathcal{I}$	0	Eichler–Brandt + local-2		A	theorem-grade
24-cell bridge	24-cell / 600-cell spectral bridge	Spectrum identification	0	Coxeter group data		A	theorem-grade
Hadronic	$r_p = 4\bar{\lambda}_p$ (proton radius)	Sets length scale	0	$\bar{\lambda}_p$ (CODATA)		B	strong
Hadronic	Neutron / pion radii from same length scale	Sets length scale	0	CODATA radii		B	strong
Hadronic	Deuteron coupling ($6 + \alpha$)	Counts cascade modes	0	α		B	strong
Cosmology	7 cosmological observables to $< 0.5\%$ of Planck 2018	Hypersphere closure	0	4 input quantities		B	strong
Atomic	$\alpha^{-1} = 137 + \pi/87$	$\pi/87$ via cascade	0	6-hypothesis stack		B	conditional
Particles	13-mass spectrum (paper-xviii)	Cascade rungs select modes	1	QCD scale		B	conditional
Cognition	Cortical phase under propofol: $t = 7.11$ (ds005620)	Closure on phase substrate	0	pre-specified analysis		C	strong
Cognition	Cross-cohort replication on ds004541: $t = 5.25$	Same substrate hypothesis	0	out-of-sample test		C	strong
Bioelectric	BETSE 6-condition planarian panel	Closure on bioelectric substrate	0	5 pre-registered predictions		C	conditional
Cognition (ARIA)	18/18 preregistered, after refinement	Closure as algorithmic kernel	0	18 prereg tests		C	strong
Biology	Volatile inhaled anaesthetics: Spearman $\rho = +0.94$ ($n = 11$)	MT-substrate hypothesis	0	exploratory, no preregistration		C	weak
Ontology	“ X exists $\iff \mathcal{C}(X) = X$ ”	Defines closure-as-existence	—	—		D	interpretive
Ontology	Recurrence: substrate as cosmological dynamic	Closure as recurrence	—	—		D	interpretive

Free-parameter count f is the number of parameters fitted against the comparison target on each row. “Fixed inputs” lists external numbers or theorems consumed by the derivation

but not fitted. Class D rows have no f value because they make no quantitative comparison. Detailed per-row commentary, baselines, and falsifiers are in the machine-readable CSV at `data/evidence_matrix.csv`.

B Claim-status table

The claim-status table records, per claim, the open bridge hypotheses on which its conditional verdict depends. The machine-readable version is at `data/claim_status_table.csv`.

Claim (short)	Open bridges	Status	Last verified
A_1 spectrum	none	closed	2026-05-26 (sim 9)
C_φ pairing 94 + 13 + 13	none	closed	2026-05-26
Universal-representation	none	closed (Eichler–Hijikata)	2026-05-26
$L(\Theta_{\mathcal{I}}, s)$ identity	none	closed	2026-05-26 (sim 13)
Proton-radius identity	H-RP (radius-rung identification)	conditional	2026-04-23
Cosmological observables	H-cosmo-closure	conditional	2026-05-18
$\alpha^{-1} = 137 + \pi/87$	6-hypothesis stack (clay-bar)	conditional	2026-04-21
13-mass spectrum	H-mass-selection	conditional	2026-04-22
Cortical phase (ds005620 / ds004541)	H-CortexClosure	open	2026-05-20
BETSE 6-condition panel	H-Bioclosure	open	2026-05-20
ARIA 18/18	H-AlgoKernel	open	2026-04-30
Volatile-anaesthetic $\rho = +0.94$	H-MT-Bridge	open (exploratory)	2026-05-24
Closure-as-existence	H-Existence (interpretation)	not assessed	—
Recurrence cosmology	H-Recurrence (interpretation)	not assessed	—

C Falsification ledger

For each row of the evidence matrix we record (i) the falsifying observation that would refute the claim, (ii) whether the falsifying observation has been attempted, and (iii) the current status. The machine-readable version is at `data/falsification_ledger.csv`.

Claim (short)	Falsifier	Attempted?	Outcome
A_1 spectrum	A 10th eigenvalue, or shifted multiplicities	yes (sim 9)	not falsified
C_φ pairing	Pairing not summing to 120, or not 94 + 13 + 13	yes (sim 9)	not falsified
Universal-representation	A totally positive prime not in image of N_H	yes (sim 9, bounded witnesses)	not falsified
$L(\Theta_{\mathcal{I}}, s)$	A coefficient disagreement at any s	yes (sim 13)	not falsified
Proton-radius	CODATA r_p deviates from $4\bar{\lambda}_p$ by $> 0.5\%$	yes (CODATA 2022)	not falsified (0.04%)
Cosmological observables	Any of $\{\Lambda, H_0, \Omega_\Lambda, \dots\}$ deviates from Planck 2018 by $> 0.5\%$	yes (Planck 2018 inferred)	not falsified

Claim (short)	Falsifier	Attempted?	Outcome
α^{-1}	CODATA α^{-1} deviates from $137 + \pi/87$ at the per-mille level	yes (CODATA 2022)	not falsified at quoted precision
Cortical phase	Out-of-cohort replication fails ($t < 2$)	yes (ds004541)	not falsified ($t = 5.25$)
BETSE 6-condition	A pre-registered prediction fails (wet-lab)	not yet attempted	pending
ARIA 18/18	A new prereg test fails	continuous	not falsified to date
Volatile $\rho = +0.94$	Replication on a new $n \geq 11$ cohort with $\rho < 0.5$	not yet attempted	pending

The ledger is updated when a falsifier is attempted or replicated. Pending entries are flagged so that no row is silently treated as confirmed when no attempt has been made.

D Glossary of bridge hypotheses

H-RP *Radius-rung identification.* The cascade rung indexing hadronic radii is identified with a specific column of the substrate. Closure depends on the column being uniquely determined by independent structural data.

H-cosmo-closure *Hypersphere closure hypothesis.* The seven joint cosmological observables are produced by the substrate’s closure dynamics with the stated input set.

H-CortexClosure *Cortical phase closure hypothesis.* The cortical phase variable analysed in Solution Lab 002 is a closure observable in the sense used in [4].

H-Bioclosure *Bioelectric closure hypothesis.* Bioelectric state evolution in planarian regeneration is a closure dynamics on the substrate.

H-AlgoKernel *ARIA algorithmic-kernel hypothesis.* The substrate kernel is the algorithmic component tested by aria’s preregistered protocol.

H-MT-Bridge *Microtubule substrate-bridge hypothesis.* The microtubule lattice is a high-coherence candidate substrate compatible with the σ -paired class structure. Note G volatile-anaesthetic finding is the exploratory empirical witness.

H-mass-selection *Mass-selection hypothesis.* The 13-mass spectrum of paper-xviii is selected by the substrate’s modal structure with one external input.

H-EulerBridge *Euler-bridge hypothesis.* The factorisation $L(\Theta_{\mathcal{I}}, s) = \zeta_K(s)\zeta_K(s-1)C_2(s)$ is the Euler-bridge form; the bridge to classical RH is *not* assumed and the programme makes no such claim.

References

- [1] Nima Arkani-Hamed and Jaroslav Trnka. The amplituhedron. *Journal of High Energy Physics*, 2014(10):30, 2014.
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- [3] Carlo Rovelli. *Quantum Gravity*. Cambridge University Press, 2004.
- [4] Lee Smart. The closure picture: How a single mathematical substrate becomes universe, life, number, and mind, 2026. Pre-peer-review preprint. <https://github.com/vfd-org/closure-picture>.